COS40007 Artificial Intelligence Engineering Applied Project Task 1

AI-Based Bridge Structural Health Monitoring System

Studio Session No: Class 4 (Tuesday 8:30am - 10:30am)
Group No: 6

Group Members :
- **Chee Chen Guo - 104829801**
- **Tevin Seneviratne - 104748003**
- **Dev Bohra - 103819241**
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Acknowledgement to Country

I acknowledge the Traditional Owners of the land on which Swinburne University of Technology's Hawthorn campus stands, the Wurundjeri people of the Kulin Nation. I pay my respects to their Elders past, present, and emerging, and recognise their continuing connection to land, waters, and community.

1. Group Agreement and Project Plan

Group style and roles

Our team decided to follow an **Agile Scrum-based approach** to manage the development of the AI-Based Bridge Structural Health Monitoring System. The project is divided into structured sprints aligned with key submission milestones, ensuring continuous progress, collaboration, and timely delivery.

Each member is assigned a **specific role based on their strengths**. Also, while maintaining shared responsibility for overall project success.

Team Members & Roles

- **Tevin Seneviratne – Scrum Master & Project Coordinator**
  Responsible for sprint planning, task allocation, tracking progress, and ensuring deadlines are met. Tevin will coordinate team activities, manage communication, and handle final submission and tutor communication.
- **Chee Chen Guo – Data Engineer**
  Responsible for data preprocessing, cleaning, handling missing values, and preparing the bridge sensor dataset for machine learning tasks.
- **Dev Bohra – Machine Learning Engineer**
  Responsible for developing, training, and evaluating machine learning models for Structural Health Index (SHI) prediction, anomaly detection, and failure probability estimation. DevOps and creating the pipeline.
- **Sandun Rajapakse– System Designer & Documentation Lead**
  Responsible for designing the system blueprint, maintaining documentation, and supporting system integration and visualisation components.

While roles are assigned based on strengths, all members will collaborate and support each other to ensure overall project success and balanced contribution.

Working Approach

- Work is divided according to roles but **reviewed collaboratively** to ensure quality.
- All key decisions, such as model selection and system architecture are made collectively.
- The Scrum Master will communicate with the tutor on behalf of the group.
- All deliverables will undergo **group review before submission**.

Timeline

The project is structured into **three realistic sprints**, aligned with submission deadlines and the AI Engineering lifecycle.

Sprint 1: Project Planning & Data Understanding (Week 4–6)

Objectives:

- Understand dataset structure and features
- Perform exploratory data analysis (EDA)
- Define project scope and problem statement
- Design system blueprint
- Establish a project plan and team agreement

Deliverables:

- Group Agreement & Project Plan
- System Blueprint Diagram
- Initial documentation

Sprint 2: Data Processing & Model Development (Week 6–9)

Objectives:

- Clean and preprocess sensor data
- Perform feature engineering
- Develop machine learning models. For example, regression for SHI prediction, classification for anomaly detection
- Conduct initial model evaluation

Deliverables:

- Preprocessed dataset
- Initial trained models
- Intermediate results for evaluation

Sprint 3: Model Optimisation, System Integration & Finalisation (Week 9–13)

Objectives:

- Optimise and fine-tune models
- Evaluate performance using appropriate metrics
- Integrate the model into a complete system pipeline
- Develop a basic dashboard or output visualisation
- Finalise report and GitHub repository

Deliverables:

- Final AI system
- Complete project report
- GitHub repository with all components

The team aims to complete all tasks **at least 2 days before deadlines** to allow time for revisions and unforeseen issues.

Communication Plan

To ensure effective collaboration, the team will follow a structured communication strategy:

Tools Used

- **Primary Communication:** Discord
- **Meetings:** Microsoft Teams, Discord, and in person
- **Collaboration & Documentation:** Google Docs
- **Version Control:** GitHub

Meeting Schedule

- Minimum **2 meetings per week**: The group will hold a meeting for Sprint planning/task allocation, which is followed by a Progress review meeting. This will help us keep track of all tasks every week.
- Additional meetings will be scheduled before major deadlines

Communication Expectations

- Members must respond within **24 hours**
- Urgent matters should be addressed within **12 hours**
- All members are expected to attend meetings or inform the team in advance

On-track behaviours

The following behaviours will contribute to successful team outcomes:

- Active participation in discussions and meetings
- On time completion of assigned tasks
- Clear, respectful, and professional communication
- Regular progress updates
- Supporting other team members when required
- Taking responsibility for assigned roles.

Off-track behaviours

The following behaviours may negatively impact team performance:

- Missing deadlines without a valid reason or communication
- Lack of contribution or engagement
- Ignoring team messages or updates
- Producing incomplete or low-quality work
- Lack of collaboration and teamwork

Resolving tensions

Conflicts will be handled using the following process to maintain a productive and respectful working environment in our group,

1. First, address the issue through open discussion within the team
2. Identify the cause and agree on a fair solution
3. If unable to resolve within the group, escalate the issue to the tutor for guidance

All members are expected to approach conflicts professionally and constructively.

Team Expectations:

1. **Each member must contribute equally and complete assigned tasks on time**
2. **Maintain consistent communication and attend scheduled meetings**
3. **Deliver high-quality work and support overall team success**

2. System Blueprint Diagram

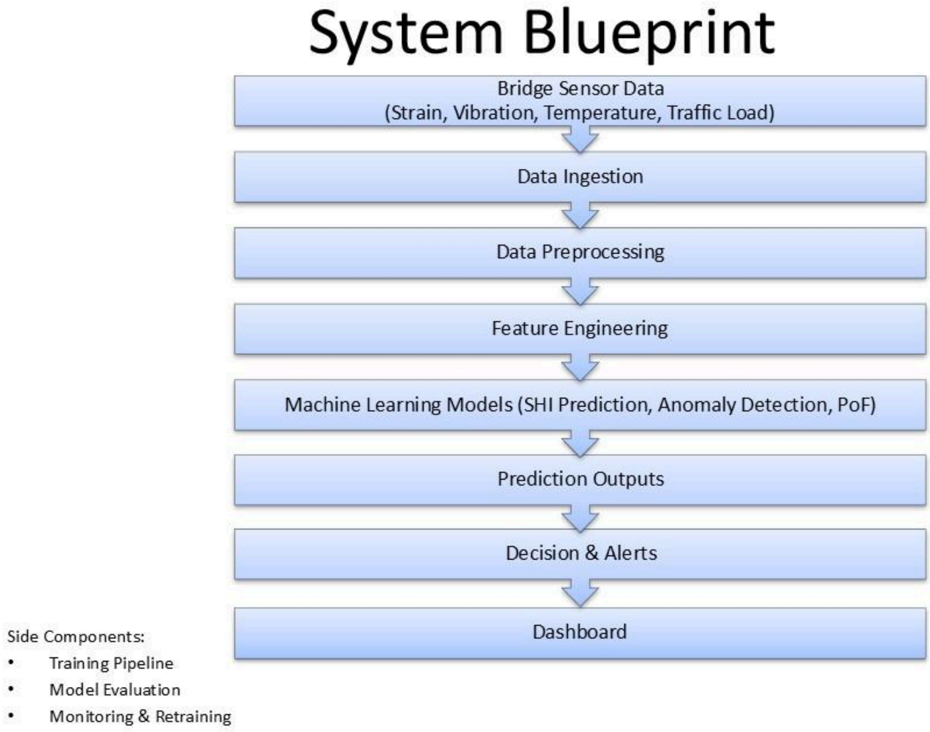

Figure 1: System blueprint diagram

The proposed AI-Based Bridge Structural Health Monitoring System is designed as a data-driven pipeline that continuously processes structural, environmental, and operational data to assess bridge health and predict potential failures.

1. Data Source & Ingestion

The system begins with a digital twin dataset representing bridge sensor readings, including structural parameters such as strain, deflection, vibration, and crack propagation, as well as environmental and load-related variables like temperature, humidity, traffic volume, and wind speed. These data are ingested from a CSV-based storage system into the processing pipeline.

2. Data Preprocessing

In this stage, raw sensor data is cleaned and prepared to ensure reliability and accuracy. This includes handling missing values, removing noise, and normalising features. Since sensor data may contain inconsistencies, preprocessing ensures that the dataset is suitable for machine learning.

3. Feature Engineering

Relevant features are selected and transformed to improve model performance. Structural indicators such as strain, vibration, and fatigue accumulation are combined with environmental factors like humidity and corrosion level, and operational factors such as traffic volume and vehicle load. This enables the model to better capture real-world bridge behaviour.

4. Machine Learning Models

The processed data is fed into machine learning models developed by the ML engineer. The system includes:

- **Regression models** for predicting Structural Health Index (SHI)
- **Classification models** for anomaly detection
- **Risk models** for Probability of Failure (PoF)

5. Prediction Outputs

The system generates key outputs including SHI predictions (current and future), anomaly detection scores, failure probabilities, and maintenance alerts. These outputs provide actionable insights into the condition of the bridge.

6. Decision and Alert System

Based on the predictions, the system categorises the structural condition into different risk levels and triggers alerts when abnormal behaviour or potential failures are detected. This enables proactive maintenance and reduces the risk of structural damage.

7. User Interface / Dashboard

The results are displayed through a user-friendly dashboard that allows engineers to monitor bridge health in real time. The dashboard includes visualisations of SHI trends, anomaly alerts, and risk indicators, supporting informed decision-making.

8. Model Training Pipeline

A separate training pipeline uses historical data to:

- Train machine learning models
- Validate and test performance
- Store trained models for deployment

9. Monitoring and Retraining

The system includes a monitoring mechanism to track model performance and detect data drift. If performance degrades or new data becomes available, the system triggers retraining to maintain accuracy and reliability.

3. MLOps & DevOps Integration

Data & Training Strategy

The dataset used in this project is a pre-collected time-series dataset stored in a CSV file, representing sensor readings from a digital twin of a smart bridge. Since the data is sequential, it will be split chronologically into training (70%), validation (15%), and testing (15%) sets to preserve temporal dependencies and avoid data leakage.

The training set will be used to learn relationships between structural, environmental, and load-related features (such as strain, crack propagation, humidity, and traffic volume) and the target outputs (e.g., Structural Health Index or risk levels). The validation set will support model tuning and hyperparameter selection, ensuring that the model generalises well rather than memorising patterns. The test set will remain completely unseen during training and will be used to provide an unbiased evaluation of model performance.

Although the current dataset is static, the system is designed with scalability in back of the head. If new data becomes available (e.g., additional time periods or simulated scenarios), it can be appended to the dataset. The model can then be retrained using the expanded dataset, allowing it to learn from more diverse conditions such as extreme weather events. This ensures that the system remains adaptable and future-ready.

Retraining Strategy

Since the dataset is not continuously updated in real time, retraining will not occur automatically. Instead, retraining will be performed when new data is introduced or when there is a need to improve model performance.

A time-based retraining strategy can be simulated by periodically retraining the model after adding new batches of data (e.g., representing new months or conditions). This ensures that the model remains aligned with the most recent data trends.

In addition, a performance-based retraining strategy will be used. If the model's performance drops below a predefined threshold (for example, reduced accuracy in classifying bridge conditions or increased prediction error for remaining useful life), retraining will be triggered. This ensures that the model remains reliable even if underlying patterns change.

Combining both strategies reflects how real-world systems operate, where models are updated both regularly and in response to performance degradation.

Monitoring & Drift Detection

Model performance will be monitored using evaluation metrics appropriate to each task. For classification tasks such as structural health prediction, metrics like accuracy, precision, and recall will be used. For regression tasks such as remaining useful life estimation, error-based metrics (e.g., mean absolute error) will be considered.

Although real-time monitoring is not possible with a static dataset, the concept can be demonstrated by evaluating the model on different time segments of the data. For example, earlier portions of the dataset can be treated as "historical data," while later portions can simulate "new incoming data."

Data drift will be detected by comparing the statistical distribution of features across these segments. Significant changes in variables such as strain, temperature, or traffic load may indicate that the operating environment has shifted.

Concept drift will be identified if the model performs well on earlier data but poorly on later data, suggesting that the relationship between inputs and outputs has changed. For example, the same strain levels may correspond to different structural conditions over time due to material fatigue.

By simulating these scenarios, the system demonstrates how monitoring and drift detection would be handled in a real-world deployment.

DevOps Practices

Version control will be managed using GitHub. All project components, including data preprocessing scripts, model training pipelines, and backend logic, will be tracked through commits and branches. A simple branching strategy (such as main, development, and feature branches) will be used to organise team contributions and minimise merge conflicts. This approach supports collaborative development, provides a clear history of changes, and allows the team to easily roll back to stable versions if needed.

A CI/CD pipeline will be conceptually implemented using GitHub Actions to introduce basic automation into the workflow. Whenever code is pushed to the repository, automated checks such as linting, basic testing, and integration validation can be triggered. This ensures that new updates do not break existing functionality. While full deployment automation is not required at this stage, the pipeline reflects modern practices and can be extended in the future to support continuous training and deployment of updated models.

Both code and models will be versioned to ensure reproducibility and traceability. Each trained model will be stored with a unique version identifier along with key metadata, such as the training data range, selected parameters, and evaluation metrics. This allows the team to compare model performance over time, track improvements, and confidently revert to a previous version if necessary. Maintaining this level of model lifecycle management ensures the system remains reliable as it evolves.

4. Project Management Setup

Trello is used to manage and track project tasks and overall progress. It organises user stories and system components into smaller, actionable tasks, which are prioritised and assigned to team members to ensure balanced workload distribution and efficient collaboration.

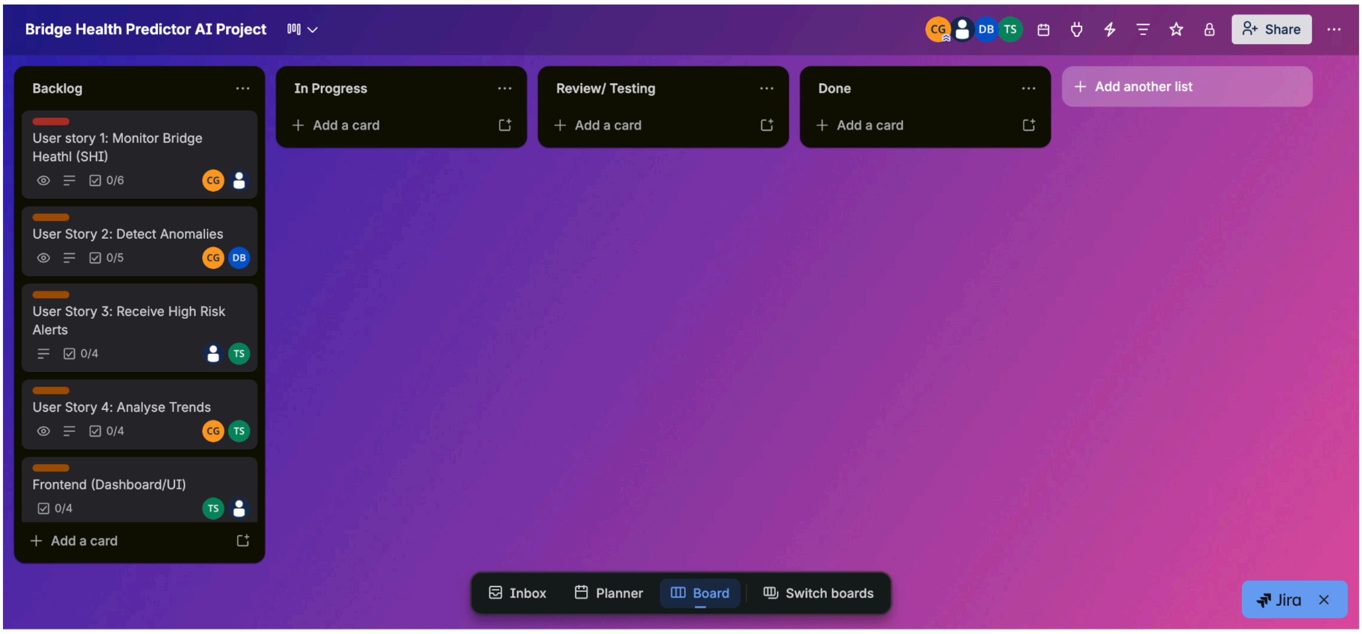

Figure 2: Image of Trello dashboard

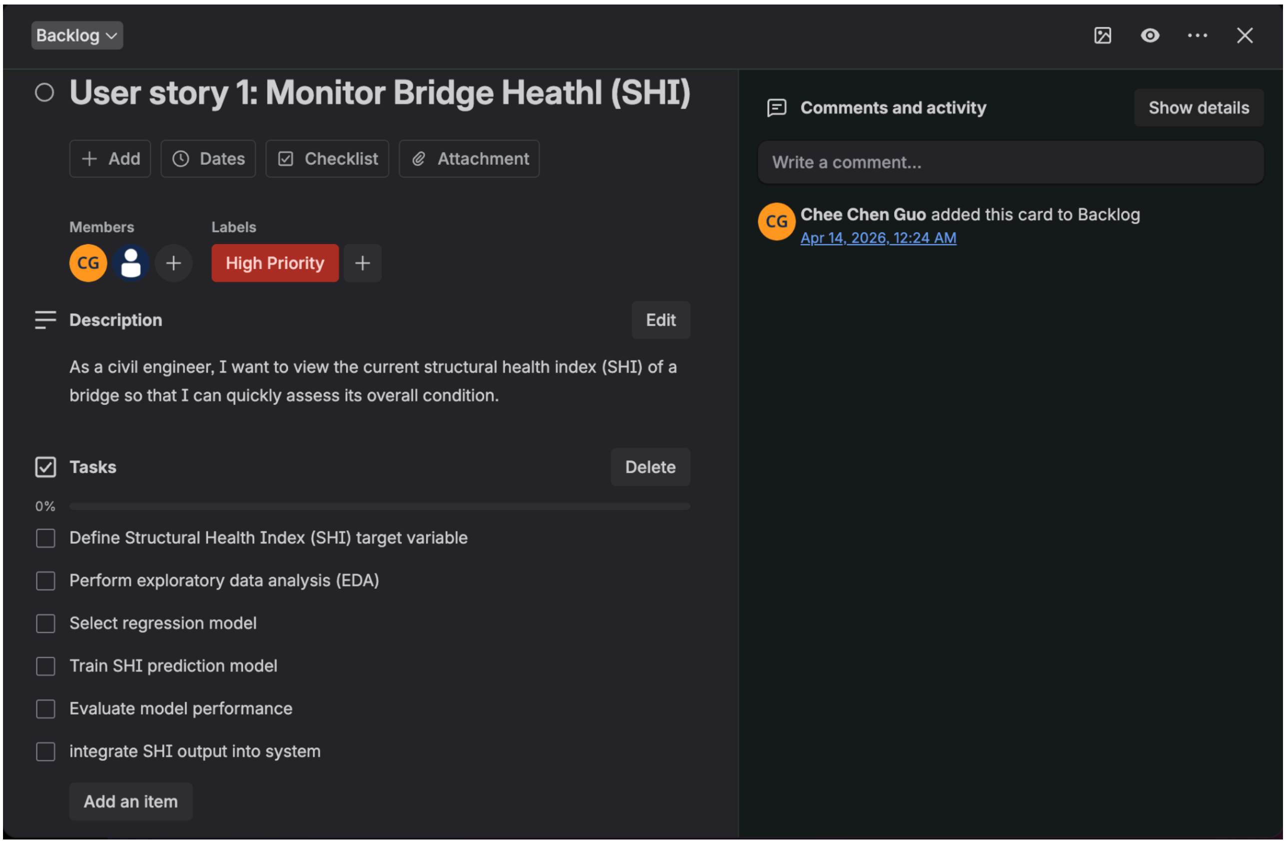

Figure 3: Image of breakdown of user story 1

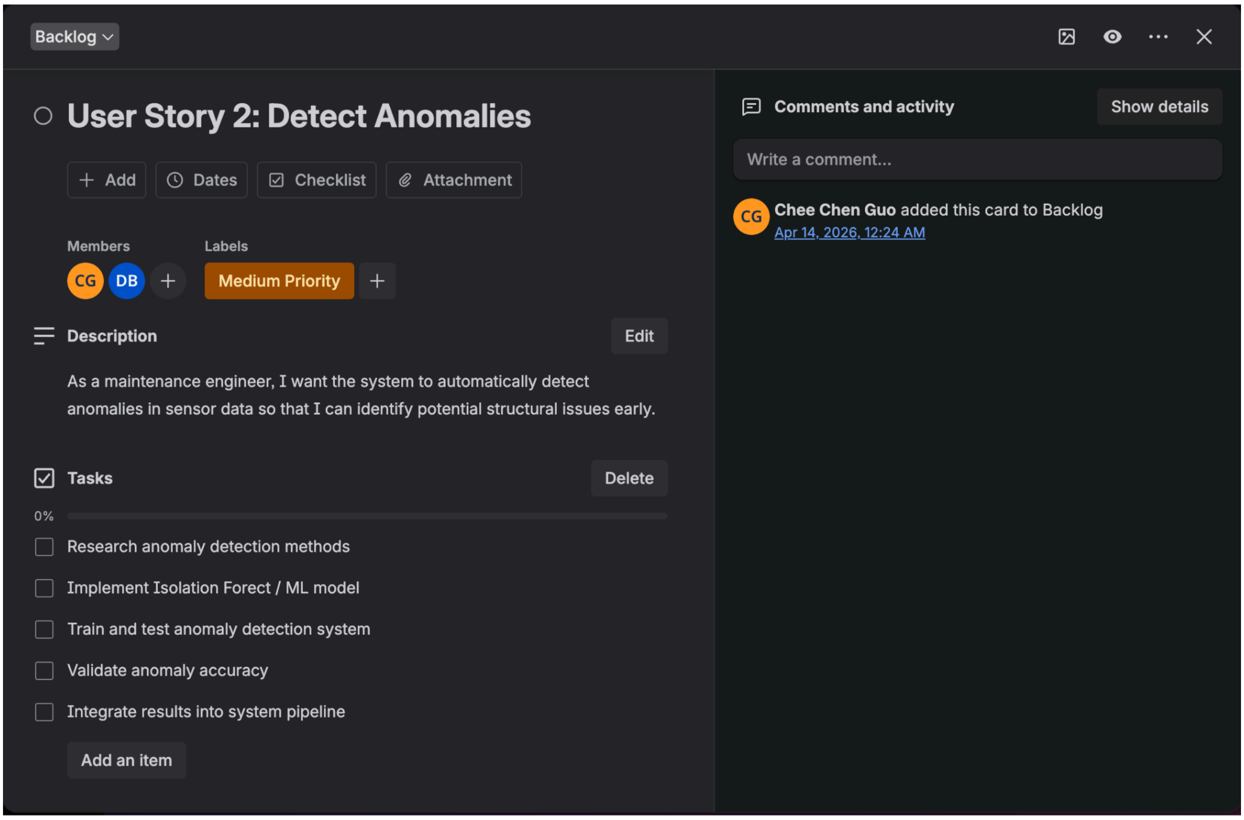

Figure 4: Image of breakdown of user story 2

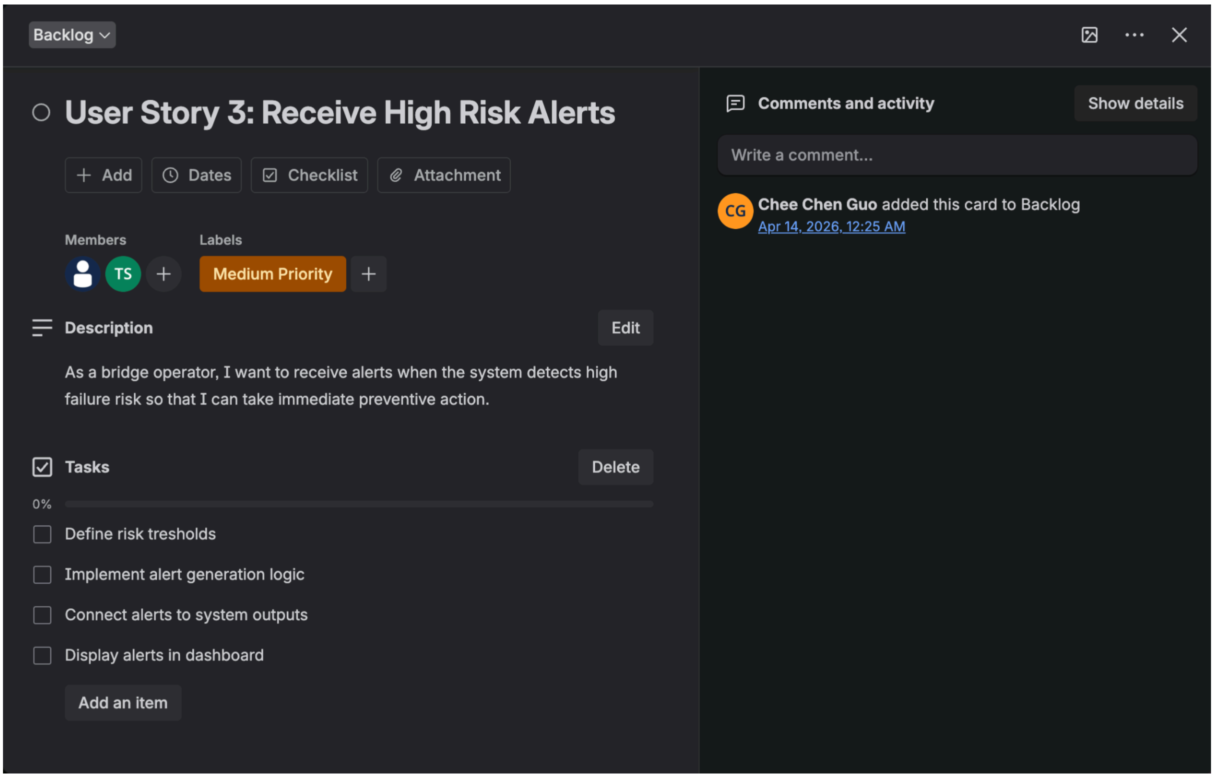

Figure 5: Image of breakdown of user story 3

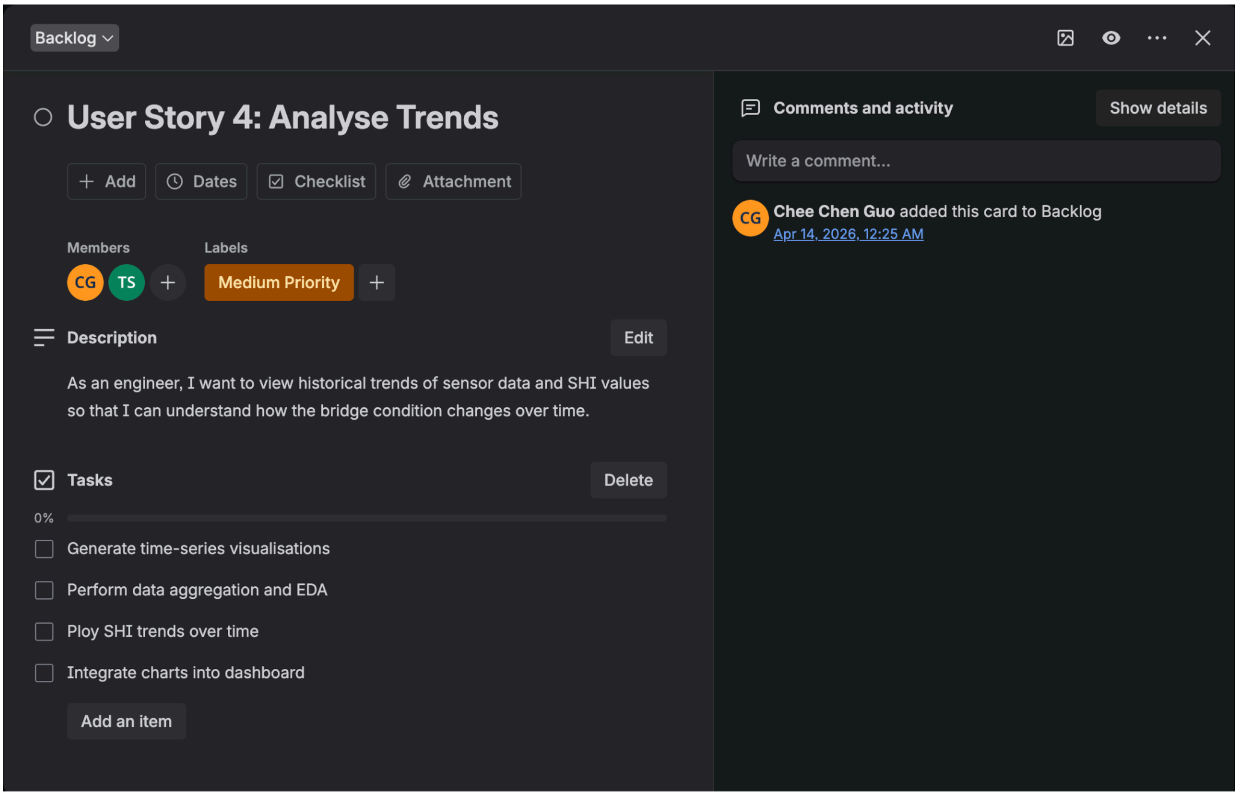

Figure 6: Image of breakdown of user story 4

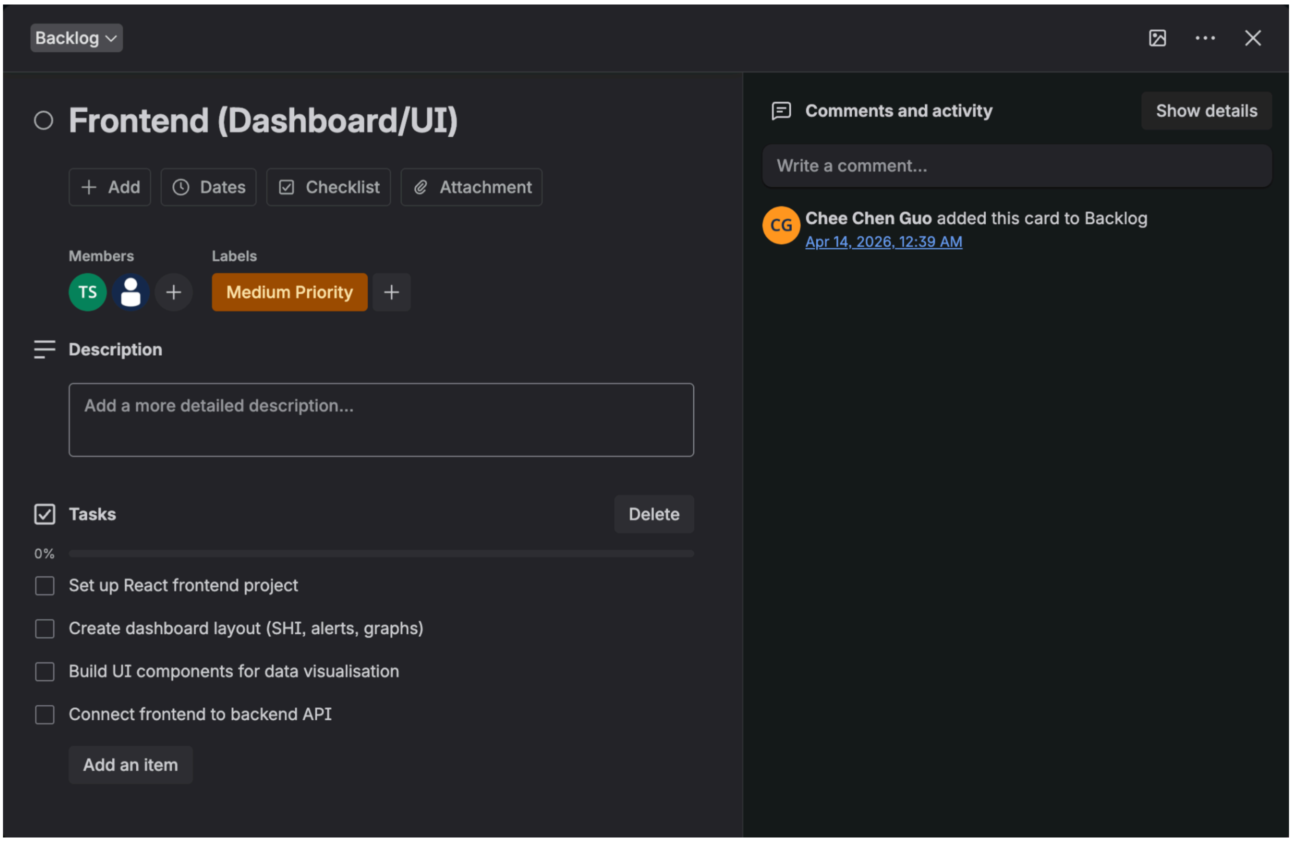

Figure 7: Image of breakdown of frontend

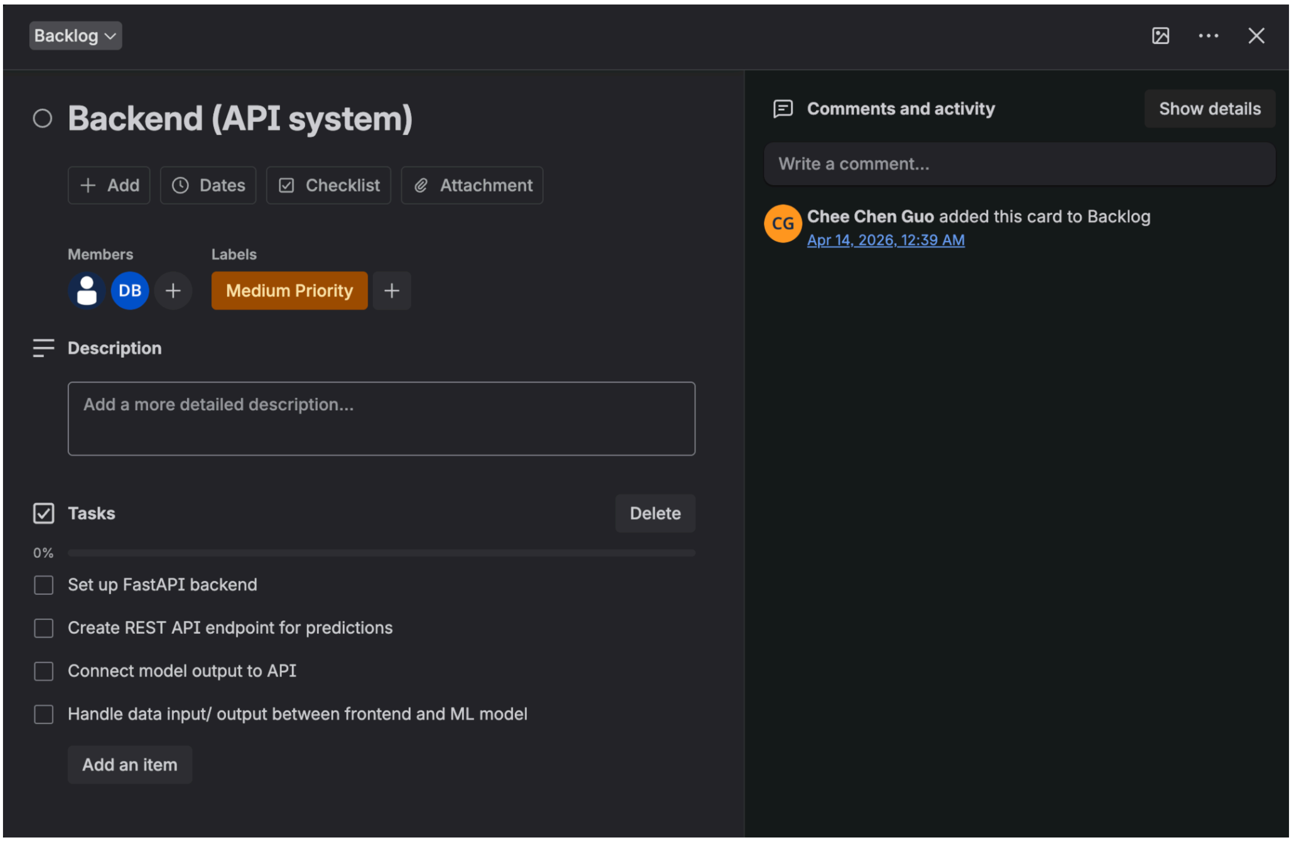

Figure 8: Image of breakdown of backend

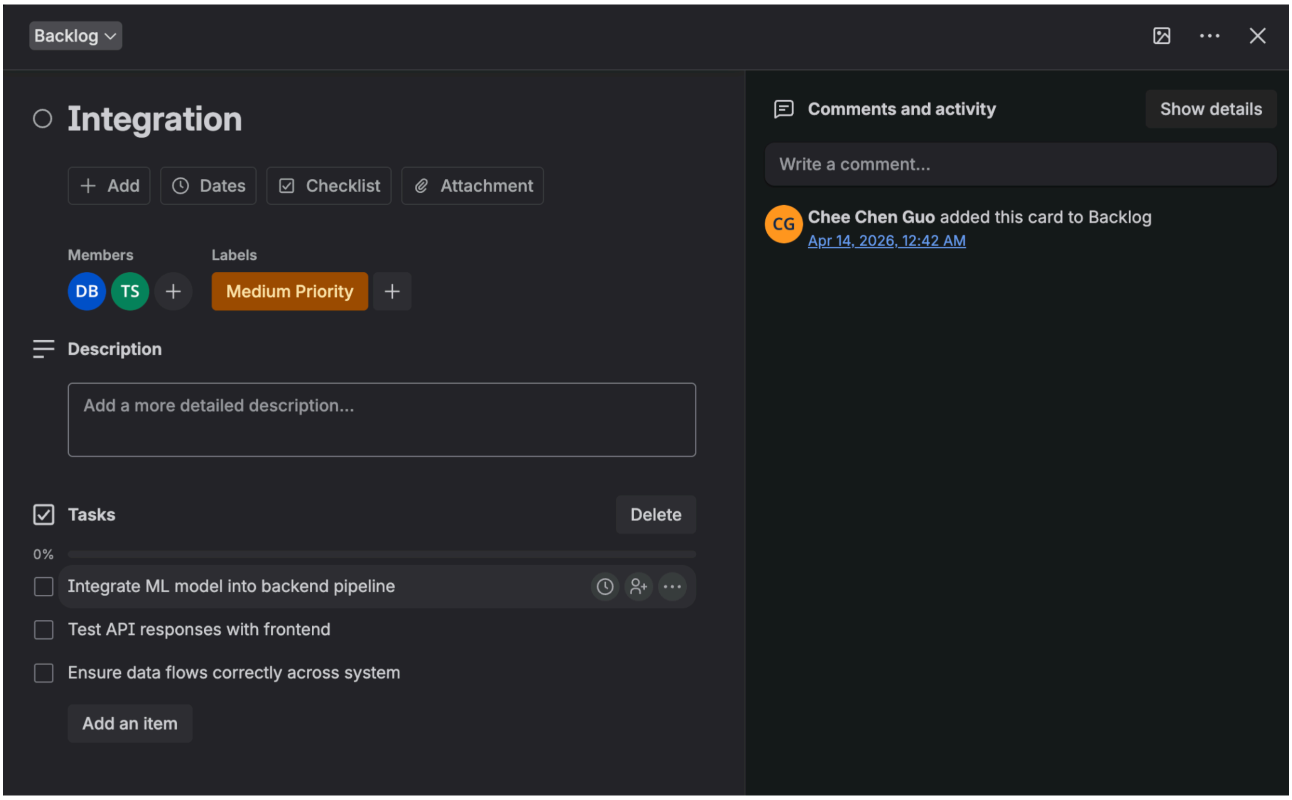

Figure 9: Image of breakdown of integration

Trello Link:
https://trello.com/invite/b/69dcfb6be309ebb25d9ecc9c/ATTIf294f19f4acf441173f8ab770df9acc294AAA074/bridge-health-predictor-ai-project

Confluence is used as a shared documentation platform to centralise all project-related information. It provides a structured space for maintaining the project description, system overview, and key design decisions, ensuring consistency, transparency, and easy access for all team members. Later on in the project, this platform can also be utilised to document information such as meeting notes, research notes, etc.

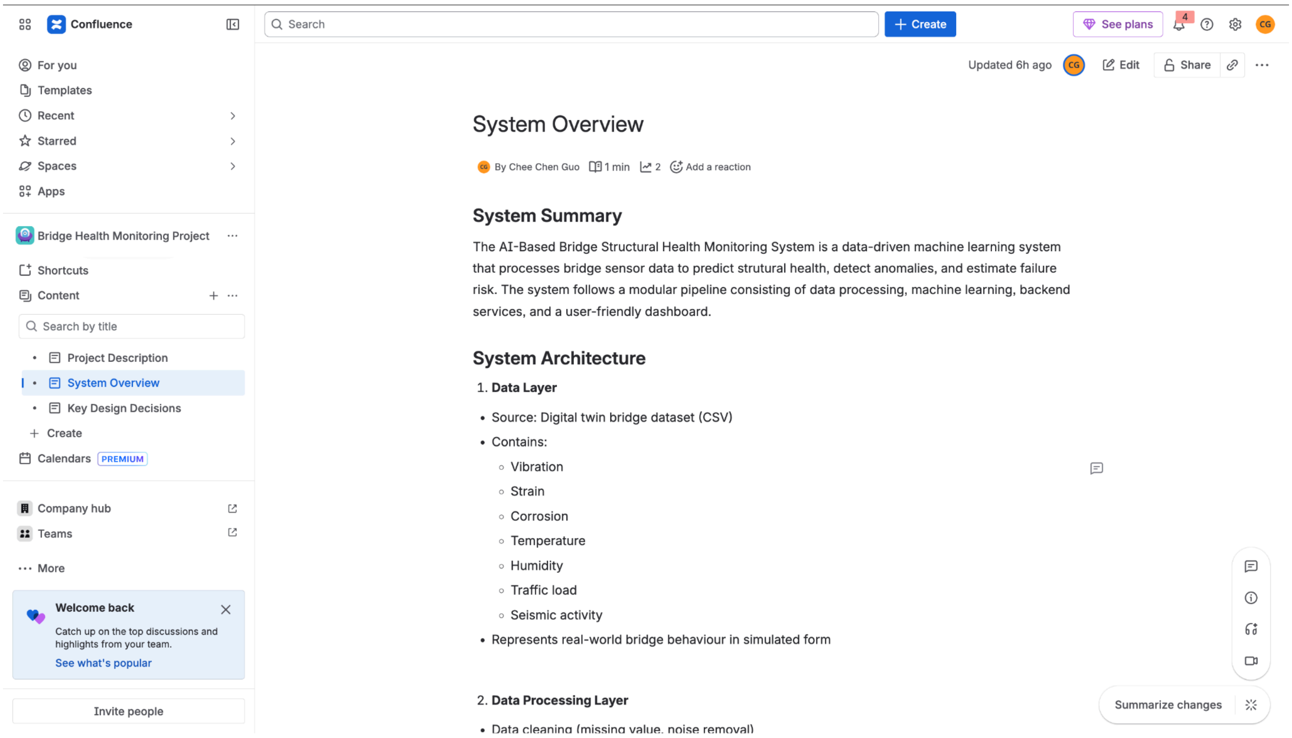

Figure 10: Image of system overview in Confluence

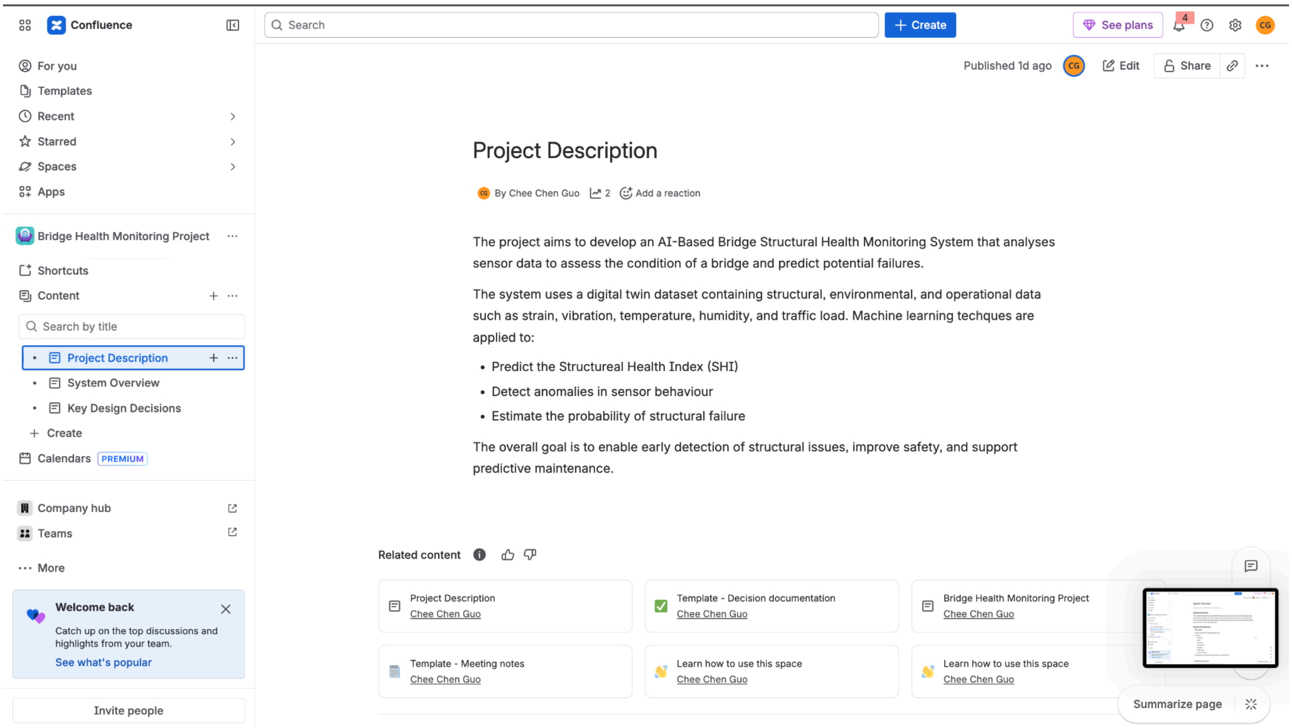

Figure 11: Image of project description in Confluence

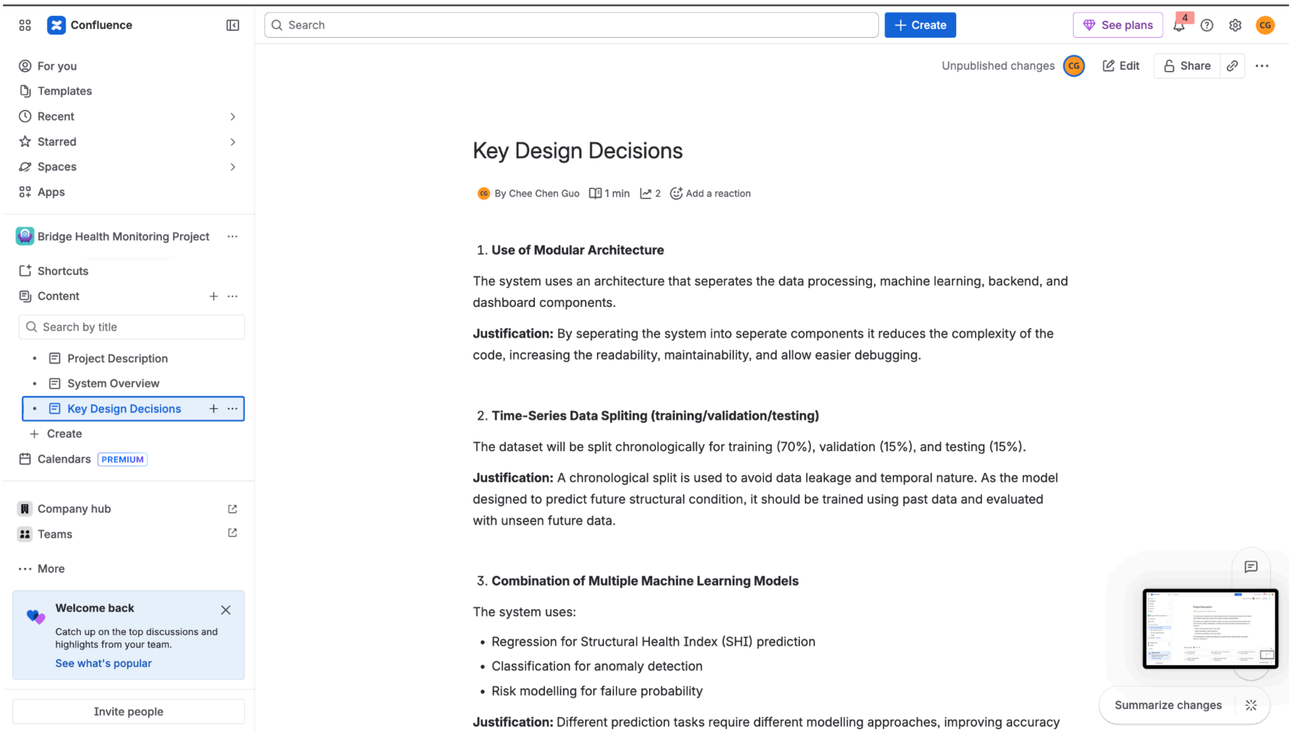

Figure 12: Image of key design decisions in Confluence

Confluence Link:
https://jackcc02.atlassian.net/wiki/x/C4AF

5. Contribution Statement

The team had meetings on Discord & also had discussions in class to decide which dataset we will be using and decided the project features & outcomes.

- Tevin Seneviratne - Finished Group Agreement & Project Plan (Section 1)
- Sandun Rajapaksa - Designed system blueprint diagram & Wrote Section 2
- Dev Bohra - Wrote Acknowledgement to Country , MLOps & DevOps Integration (Section 3) & Contribution Statement (Section 5)
- Chen Chen Guo - Completed Project Management Setup (Section 4) including setting up Trello & Confluence

6. GenAI Declaration

Generative AI tools were used to support the comparison of different project options by providing a second opinion on their strengths, difficulty, and overall potential. The AI assisted in structuring the comparison and suggesting evaluation criteria such as machine learning depth, visual appeal, and uniqueness.

The final comparison, including the ratings and conclusions, was based on the team's own understanding and decision-making. The AI was used only as a supporting tool to help guide discussion and refine ideas, rather than to replace independent analysis.

⚖️ Honest comparison (ALL 3 top options)

Factor	UR3 Cobot	Roof CV	Bridge Digital Twin
ML Depth	⭐⭐⭐⭐⭐	⭐⭐⭐	⭐⭐⭐⭐⭐
Visual Appeal	⭐⭐	⭐⭐⭐⭐⭐	⭐⭐⭐⭐
Uniqueness	⭐⭐⭐⭐	⭐⭐	⭐⭐⭐⭐⭐
Difficulty	High	Medium	Medium-High
Portfolio Value	⭐⭐⭐⭐⭐	⭐⭐⭐⭐	⭐⭐⭐⭐⭐

Figure 13: GenAI's second opinion on shortlisted project datasets